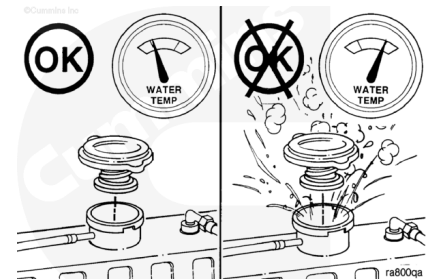


Trainings ▾

Test

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

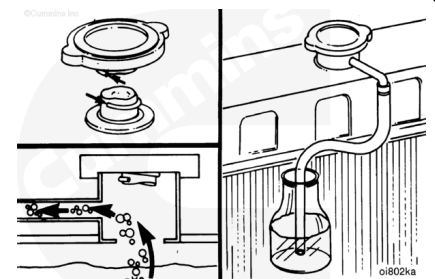


Allow the engine to cool. Remove the radiator cap.

The pressure cap **must** make a tight seal.

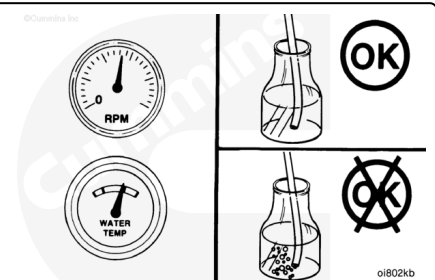
Install a radiator pressure cap that has had the spring and pressure relief valve removed.

Attach a rubber hose to the radiator overflow connection. Put the free end of the hose in a container of water.



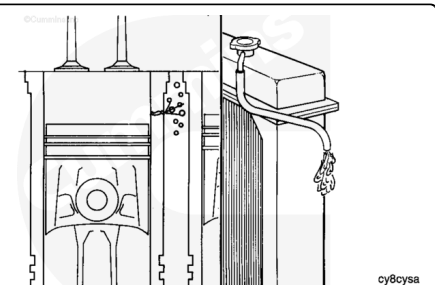
Operate the engine at rated rpm until it reaches a temperature of 80°C [180°F] with the thermostat open.

Check for a continuous flow of air bubbles from the hose in the water container.



If a flow of air bubbles exist, check for defects in the following locations.

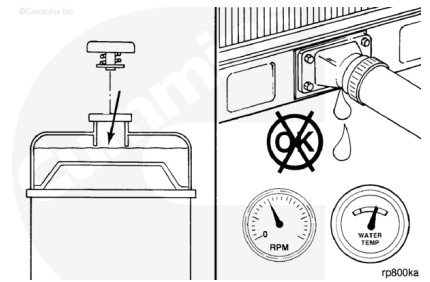
- A cylinder head combustion gas leak. Inspect the cylinder head and gasket for damage. Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/89/89-002-004.html).
- Incorrect cylinder liner protrusion. Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/89/89-001-028.html).



- Cracked or broken cylinder liners. Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/89/89-001-028.html).

If no air is in the system, do the following:

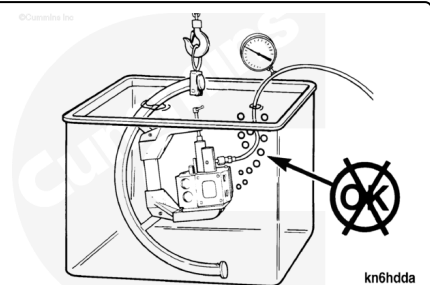
- Remove the test equipment.
- Check the coolant level and fill, if necessary.
- Install the correct radiator pressure cap.
- Operate the engine to a temperature of 80°C [180°F], and check for coolant leaks.



Cylinder Head and Cylinder Liner

A cracked or defective cylinder head will pressurize the cooling system and force the coolant out the overflow tube.

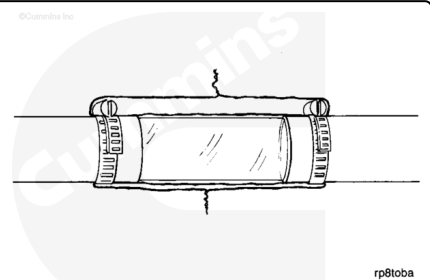
Check the cylinder head gasket. The part **must** be replaced if defective. Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/89/89-002-004.html) or Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/89/89-001-028.html).



Sight Glass Method

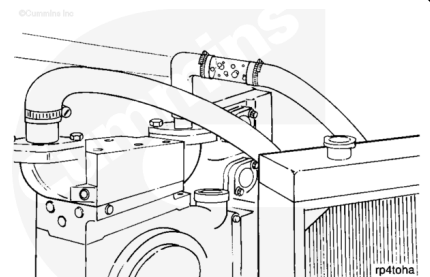
Put a clear Pyrex® tube in the water connection between the engine radiator and the thermostat housing.

If desired, put the Pyrex® tube in the vent line between the thermostat housing and the radiator.



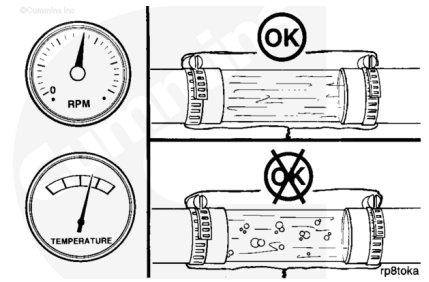
⚠ CAUTION ⚠

Use wire lace to connect the tube and hoses. Personal injury can result if the Pyrex® tube separates from the hoses.



Operate the engine at 75 percent of rated rpm until it reaches an operating temperature of 80°C [180°F].

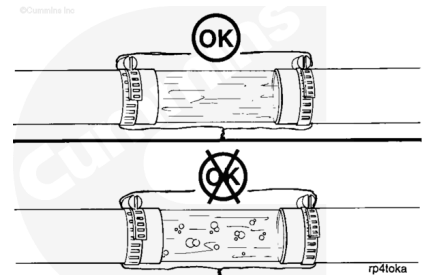
Check for a continuous flow of air bubbles in the hose.



A continuous flow of air bubbles indicates excessive air in the system. Continue to operate the engine for 25 minutes.

If the flow of bubbles does **not** stop, check for the following:

- A cylinder head combustion gas leak. Inspect the cylinder head and gasket for damage. Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/89/89-002-004.html).
- Incorrect cylinder liner protrusion. Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/89/89-001-028.html).
- Cracked or broken cylinder liners. Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/89/89-001-028.html).



If no air is found in the system or if the flow of bubbles stops within 25 minutes, do the following.

- Remove the test equipment.
- Check the coolant level and fill if necessary.
- Install the correct radiator pressure cap.
- Operate the engine until it reaches a temperature of 70°C [160°F] and check for coolant leaks.

